

The key advantages of using a simulation based framework in cloud computing are listed below:

- Simulators facilitate dynamic and flexible configuration and development environments. They enable researchers cum system administrators to learn and make the real-time cloud environment more scalable and reliable.
- Simulators provide easy-to-use command/graphical interfaces with lots of customisation options that help researchers visualise real world scenarios, thus avoiding all sorts of real-time problems.
- Simulators are the most cost-effective solution for real world implementation, as they are first designed, then developed, properly tested and if required, even redesigned, rebuilt and retested. Doing this in the real world is quite challenging in terms of cost and time.
- Simulators also enable researchers to simulate cloud environments with their own proposed performance, security and other provisioning algorithms.

Some of the top free and open source cloud computing simulators available today are explored in this article.

CloudSim

CloudSim is a new, highly generalised and extensible Java based simulation tool kit, and is actually regarded as a software framework. It supports several core functionalities like queuing and processing of events, the creation of CloudSim entities, communication among components and the management of the simulation clock. CloudSim has been developed by the CLOUDS Laboratory of the Computer Science and Software Engineering Department of the University of Melbourne, Australia by Prof. (Dr) Rajkumar Buyya. This tool kit enables seamless modelling, simulation and experimentation in cloud computing and application services. It can be termed as 'running a model of an environment by taking the hardware as base, where technology-specific details are abstracted.'

CloudSim features include basic classes for deriving data centres, virtual machines, applications, users, computational resources, and policies for managing diverse parts of the system like scheduling and provisioning. It implements general application provisioning techniques, which can be extended easily with minimal effort.

The CloudSim simulator example depicting the creation of a data centre with one host running one cloudlet on it is demonstrated in Figure 1.

The latest version of CloudSim is 4.0. Its features are listed below:

- Supports modelling and simulation of large scale cloud computing data centres.
- Supports modelling and simulation of virtualised server hosts, along with customisable policies for provisioning host resources to virtual machines.
- Supports dynamic inclusion of simulation elements, discontinuations and restarts.

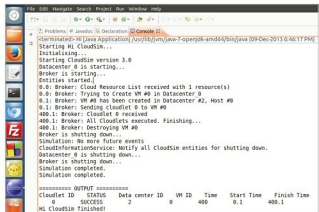


Figure 1: CloudSim live simulation

- Has support for user defined policies for allocating hosts to virtual machines (VMs).
- Supports the creation of various data centre network topologies, message-passing applications and energy-aware computational resources.
- Has the capability to simulate a federated cloud environment that inter-networks resources from both private and public domains. This is a critical feature for research into cloudbursts and automatic application scaling.

CloudAnalyst

CloudAnalyst, a GUI based simulator derived from CloudSim, has some extended features and capabilities. CloudAnalyst was proposed by Bhatiya Wickremasinghe and Rajkumar Buyya at the CLOUDS Laboratory of the Computer Science and Software Engineering Department of the University of Melbourne, Australia. The simulator supports the evaluation of social network tools according to the geographical distribution of users and data centres. It can be applied to determine the behaviour of large scale Internet applications in the cloud, and also enables a modeller for looping simulations and to conduct a series of simulations with slight variations in parameters.

CloudAnalyst is regarded as a powerful simulation framework for deploying real-time data centres and monitoring load balancing, cloud cluster monitoring and



Figure 2: GUI of CloudAnalyst